KATHERINE LALIOTIS

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EDUCATION

The Ohio State University — Columbus, OH

- Doctorate of Philosophy (PhD), Physics; Adviser: Christopher M. Hirata — *Expected 2026*
  Thesis: Optimizing Joint Processing for Stage IV Weak Gravitational Lensing Surveys
- Master's of Science (MS), Physics; Adviser: Christopher M. Hirata — December 2023

Whitman College — Walla Walla, WA

- Bachelor of Arts in Physics & Astronomy, Minor in Mathematics — May 2021
  Honors: Magna Cum Laude, Honors in Major, Walter A. Brattain Scholarship

HONORS AND AWARDS

- Presidential Fellowship, Ohio State University — 2025–2026
- Dept. of Energy Office of Science Graduate Student Research Fellowship — 2024–2025
- Physics Service Courses Outstanding Graduate Teaching Assistant Award — 2024
- Physics Departmental Service Award, for service on Climate and Diversity Committee — 2024
- Honorable Mention, NSF Graduate Research Fellowships Program — 2022
- University Fellowship, the Ohio State University — 2021–2022

FIRST AUTHOR PUBLICATIONS

- **Laliotis, K.**, Hirata, C., Macbeth, E., et al. "Removing correlated noise from the Nancy Grace Roman Space Telescope sensor chip array through destriping algorithms." Submitted to PASJ. Manuscript available on Arxiv.
- **Laliotis, Katherine**, et al. "Analysis of Biasing from Noise from the Nancy Grace Roman Space Telescope: Implications for Weak Lensing." Publications of the Astronomical Society of the Pacific, vol. 136, no. 12, 1 Dec. 2024, p. 124506, https://doi.org/10.1088/1538-3873/ad9bec.
- **Laliotis, Katherine**, Burt, J. A., Mamajek, E. E., Li, Z., Perdelwitz, V., et al. (2023). Doppler constraints on planetary companions to nearby sun-like stars: An archival radial velocity survey of southern targets for proposed NASA Direct Imaging Missions. The Astronomical Journal, 165(4), 176. https://doi.org/10.3847/1538-3881/acc067

CO-AUTHORED PUBLICATIONS

- Cao, K., Hirata, C., **Laliotis, K.**, Yamamoto, M., Macbeth, E., & Troxel, M. (2025). Simulating Image Coaddition with the Nancy Grace Roman Space Telescope. IV. Hyperparameter Optimization and Experimental Features. arXiv e-prints, arXiv:2509.18286.

- OpenUniverse, et al. "OpenUniverse2024: A Shared, Simulated View of the Sky for the next Generation of Cosmological Surveys." ArXiv.org, 2024, arxiv.org/abs/2501.05632.

- Cao, K., Hirata, C., **Laliotis, K.**, Yamamoto, M., Macbeth, E., & Troxel, M. (2025). Simulating Image Coaddition with the Nancy Grace Roman Space Telescope. III. Software Improvements and New Linear Algebra Strategies. APJS, 277(2), 55.

- Yamamoto, M., **Laliotis, K.**, Macbeth, E., Zhang, T., Hirata, C., et al. (2024). Simulating image coaddition with the Nancy Grace Roman Space Telescope - II. Analysis of the simulated images and implications for weak lensing. MNRAS, 528(4), 6680-6705.

- Hirata, C., Yamamoto, M., **Laliotis, K.**, Macbeth, E., Troxel, M., et al. (2024). Simulating image coaddition with the Nancy Grace Roman Space Telescope - I. Simulation methodology and general results. MNRAS, 528(2), 2533-2561.

PUBLICATIONS IN PREPARATION

- **Laliotis, K.**, Leget, P.F., Roodman, A., et al. "Impacts of CCD Non-Flatness in Rubin LSSTCam PSF Measurements and Modeling" To be posted in Rubin Tech Notes. Draft Rubin TechNote on GitHub.

PUBLIC WRITING

- **Laliotis, Katherine.** "Europe's Extremely Large Telescope Faces a New Dire Threat." Scientific American, 30 Jan. 2025, www.scientificamerican.com/article/europes-extremely-large-telescope-faces-a-new-dire-threat/.

- **Laliotis, Katherine.** "Saving the Chandra X-Ray Observatory." Undark Magazine, 19 Sept. 2024, undark.org/2024/09/19/opinion-saving-chandra-x-ray-observatory/.

PRESENTATIONS & POSTERS

Conferences & Collaborations

- Roman High Latitude Imaging Survey (HLIS) PIT In-Person Meeting, UPenn — Nov. 2025
  Presentation: Suppressing Correlated Noise Stripes with imDestripe for Precise Weak Lensing with Roman

- NASA Physics of the Cosmos Early Career Workshop — Sept. 2025
  Presentation: Characterizing and Mitigating Noise on Roman Space Telescope Detectors for Precise Weak Gravitational Lensing

- Cosmic Cartography with Roman Conference, STScI — July 2025
  Poster: Removing correlated noise from the Roman sensor chip array through destriping algorithms

- Roman High Latitude Imaging Survey (HLIS) PIT In-Person Meeting, IPAC/CalTech — Oct. 2024
  Presentation: Analysis of biasing from noise on the Nancy Grace Roman Space Telescope and Implications for Weak Lensing

- Image Sensors for Precision Astronomy, SLAC — March 2024
  Poster: Read Noise Biasing on the Nancy Grace Roman Space Telescope

Public Engagement

- Speaker Series, San Mateo County Astronomical Society — Oct. 2025
  Lecture: Illuminating the Universe Through Weak Gravitational Lensing

TEACHING AND MENTORSHIP

- Mentored and co-advised undergraduate researchers in the Hirata group at OSU:
  Emily Macbeth — August 2023 - May 2025
  Amy Albert — August 2025 - Present

- Academic Facilitator: Undergraduate Residential Summer Access (URSA) Program, Summer 2024
  Designed and taught a 2-week group-work centered curriculum exploring the question "What is Time?" through the lens of astrophysics for incoming OSU physics and astronomy students from underrepresented backgrounds in STEM.

- Teaching Assistant, OSU Physics, Fall 2022 & Spring 2023
  Taught two sections of 30 students in twice weekly recitations and weekly labs; held twice-weekly office hours for students; and designed study guides and practice problems for students before exams.

LEADERSHIP & OUTREACH

Roman HLIS Project Infrastructure Team

Co-Chair, Junior Members of the PIT (JuMP) — *May 2025 - Present*

- Created the HLIS PIT Mentorship program, and launched the first cohort
- Organized 2 in-person JuMP events
- Organized 2 virtual JuMP events, including initiating a regular lecture series

News Coordinator — *Sept. 2024 - Present*

- Manage the News page on the PIT website; solicited 2 articles from contributing members and wrote 3 articles

Polaris Program, the Ohio State University — April 2022 - November 2024
Parliamentarian

- Planned and facilitated weekly meetings of the 12-person Polaris organization Leadership Team of grads and undergrads across different subfields of physics and astronomy
- Created and documented new Recruitment & Marketing, Grants, and High School Outreach committees
- Presented a twice-yearly report to community stakeholders detailing the use of our $60,000 budget, the successes of our outreach programs, and ideal areas of prospective growth

URSA **Polaris Program, the Ohio State University**
General Facilitator — *Summer 2022*

- Marketed URSA to over 100 incoming physics/astronomy students from underrepresented backgrounds via mail, email, phone, and social media
- Organized logistics of the 15-student program, including housing, meals, transportation, technology access, classroom spaces, and communication channels between facilitators and participants
- Designed and facilitated daily community-building social activities aimed at fostering connections between the 15-student cohort